

Latest in Post-Quantum Cryptography Migration -- diff report

Generated: 2026-09-07T19:24:57.496965 | **Period:** since 2026-08-29 | **Leaves with news:** 64 of 99

Fresh developments: 59 (as_of within 14 days of the window) | **Superseded/conflicting:** 25 | **Background captured:** 83 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 26.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
Attestation protocols & silicon RoT	5	1	0	8
Platform code signing	3	0	0	0
Public CA ecosystem	2	0	1	0
IPsec & VPN protocols	3	0	0	3
QRNG & QKD adjacency	3	0	0	1
GNSS & satellite links	3	0	0	4
Interop & test tooling	3	0	0	0
PKCS#11 and provider plumbing	1	1	0	2
OTA update signing	2	0	0	4
Language runtime TLS surfaces	0	2	0	2
Enterprise CA platforms	1	1	0	0
Certificate formats & hybrid constructions	2	0	0	0
CDN & edge	2	0	0	1
SASE & zero trust	0	0	2	1
Key escrow, archive & re-encryption	2	0	0	1
ADCs & load balancers	2	0	0	1
Routers, switches, SD-WAN	2	0	0	0
Backup & storage systems	1	1	0	0
Peripherals & physical access	2	0	0	1
Payments hardware	1	1	0	0
Automotive	2	0	0	2
EU & UK guidance	1	1	0	2
Procurement, contract language & audit	2	0	0	0
CPU & ISA extensions	1	0	1	1
OpenSSL and forks	0	0	1	1

Linux distributions & kernel crypto	0	1	0	0
Database & messaging drivers	1	0	0	0
Container signing & CI/CD	0	1	0	0
Enterprise auth protocols	0	1	0	1
Passwordless & hardware tokens	1	0	0	0
Revocation & status infrastructure	0	1	0	0
Long-term signature validation & document signing	1	0	0	0
TLS 1.3 hybrid key exchange	1	0	0	0
QUIC, HTTP/3, DTLS	1	0	0	2
Link & access layer	1	0	0	0
Secure messaging protocols	1	0	0	0
HSM vendors	0	0	1	2
VPN concentrators & WAN encryptors	1	0	0	0
Disk & data-at-rest encryption	1	0	0	0
Industrial control protocols	0	0	1	1
Energy & grid	1	0	0	0
Physical security devices	1	0	0	3
Medical devices	0	1	0	1
Short-range IoT	1	0	0	0
Crypto discovery & CBOM	0	1	0	2
Performance & side-channel tooling	0	1	0	3
Certification schemes	1	0	0	0
SWIFT & ISO 20022	0	1	0	2
ERP & business platforms	0	1	0	1
VDI & remote access	0	0	1	1
Language-native crypto stacks	0	0	0	2
Embedded & alternative libraries	0	0	0	5
Apple platforms	0	0	0	3
TPM & TCG specifications	0	0	0	2
Artifact signing frameworks	0	0	0	3
Container runtimes	0	0	0	1
Service mesh & workload identity	0	0	0	3
Identity provider platforms	0	0	0	1
SSH	0	0	0	2

DNS & time	0	0	0	2
Email security	0	0	0	1
Confidential computing attestation	0	0	0	1
Sector compliance regimes	0	0	0	1
Financial standards bodies	0	0	0	2

Firmware & root of trust

Attestation protocols & silicon RoT (~~attestation-protocols~~)

New developments (5)

Caliptra

- First Caliptra 2.1 Subsystem silicon has successfully powered on as of August 19, 2026. ([Caliptra Blog](#), as of 2026-08-19, confidence: medium, source: grok)
- *New silicon-bring-up milestone for Caliptra 2.1 not previously recorded.*

OpenTitan

- OpenTitan introduces new top-level design 'Earl Grey 2' with enhanced hardened PQC support (ML-KEM, ML-DSA), CHERI memory safety, and RRAM, with initial implementation targeted for 2026 Q1-Q3. ([OpenTitan Roadmap](#), as of 2026-08-25, confidence: high, source: grok)
- *New OpenTitan top-level design (Earl Grey 2) not previously present in the knowledge base.*
- OpenTitan project announces new 'Peppermint' integrated root-of-trust top-level optimized for area-constrained SoCs (wearables, IoT), incorporating PQC support, with design in H2 2026 and completion in H1 2027. ([OpenTitan project kicks off the Peppermint integrated root-of-trust for area-constrained SoCs](#), as of 2026-08-25, confidence: medium, source: grok)
- *New OpenTitan top-level design (Peppermint) not previously present in the knowledge base.*

SPDM

- DMTF SPDM standards page (updated 2026-08-28) lists SPDM 1.4.1 as published July 2026 and an upcoming SPDM Authorization White Paper targeted for 2026Q3. ([SPDM | DMTF](#), as of 2026-08-28, confidence: high, source: grok)
- *New information about SPDM 1.4.1 publication and an upcoming Authorization White Paper, not previously present.*

measured boot

- TCG issued guidance defining requirements for PQC-ready TPMs, identifying PC Client Platform TPM Profile (PTP) 1.07 as the baseline and establishing separate 'PQC-ready' and 'PQC-upgradable' designations based on TPM 2.0 Library v1.85 PQC support. ([TCG Defines Requirements for PQC-Ready Trusted Platform Modules](#), as of 2026-08-24, confidence: medium, source: grok)
- *New TCG certification/designation framework (PTP 1.07 baseline, PQC-ready vs PQC-upgradable) not previously recorded in the knowledge base.*

Superseded (1)

SPDM

- libspdm reference implementation reached version 3.8.2 as of April 2026. ([SPDM | DMTF](#), as of 2026-08-28, confidence: high, source: grok)
- supersedes: "[d0002:3] libspdm 3.8 was released, including an update to OpenSSL 3.5.1 which adds PQC algorithm support"
- *Reports a newer point release (3.8.2) of libspdm beyond the previously recorded 3.8 release.*

Newly captured background -- older material, first time in the knowledge base (8)

Caliptra

- Caliptra implements a dual-signature scheme using ECDSA P-384 and ML-DSA-87 for authenticating firmware images (including FMC and Runtime layers) and generating dual-signed certificates in its layered DICE identity architecture. ([Caliptra.md](#), as of undated, confidence: high, source: grok)
- *Elaborates on the previously recorded dual ML-DSA-87/ECDSA scheme [d0001:8] with new specifics (P-384 curve, FMC/Runtime firmware layers, dual-signed certificates, DICE identity architecture) not stated before.*
- Caliptra v2.x FIPS certification of cryptographic algorithms completed as of July 17, 2026. ([Caliptra Blog](#), as of 2026-07-17, confidence: medium, source: grok)
- *New certification-completion milestone for Caliptra v2.x not previously recorded.*

OTBN hardware extensions for ML-DSA and ML-KEM — approved, landing in 2026

- OpenTitan partners have approved hardware extensions to the OpenTitan Big Number accelerator (OTBN) to enable ML-DSA (FIPS 204) and ML-KEM (FIPS 203) implementations: expanded data memory (4 KiB to 32 KiB), a direct interface to the side-channel-hardened KMAC block, a 32-bit SIMD ISA extension for polynomial arithmetic/NTT acceleration, and automated hardware mask conversion between masking schemes. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *First mention of OTBN hardware extensions for ML-DSA/ML-KEM in the knowledge base.*
- The mask-conversion hardware exists because side-channel hardening otherwise dominates cost: lowRISC cites masking runtime overhead that could reach roughly 90% without it; the stated motivation for the whole extension set is side-channel-resilient PQC signing (not just verification) on the root of trust. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *New engineering rationale/detail not previously recorded, distinct from the general extension description.*
- The hardware changes are planned to complete in 2026, with work already underway in the OpenTitan GitHub master branch as of February 2026; contributors named include Google, Nuvoton, and academic teams including ETH Zurich's Integrated Systems Laboratory. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *New timeline/contributor detail not previously present.*

OpenTitan Earlgrey — shipped SLH-DSA secure boot

- OpenTitan's first production silicon, the "Earlgrey" chip, supports hardware-accelerated SPHINCS+ signature verification for secure boot, and the implementation is compliant with FIPS 205 SLH-DSA as standardized by NIST in 2024. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *No prior claim in the base or deltas covers OpenTitan/Earlgrey's SLH-DSA secure-boot verification capability.*
- Mass production of Earlgrey by Taiwanese manufacturer Nuvoton began in February 2025 per Google's announcement, with the chips slated for use as the hardware root of trust in new Chromebooks and Google's data-center servers, meaning the SLH-DSA-verified boot path is shipping in consumer and infrastructure hardware, not sitting in a reference design. ([heise.de — Open and quantum safe RISC V security chip OpenTitan for Chromebooks 10274833](#), as of 2025-02-28, confidence: medium, source: ingest)

- *New deployment/production detail not previously present in the knowledge base.*

ZeroRISC commercial stack

- ZeroRISC, with the Max Planck Institute for Security and Privacy and Academia Sinica, announced on 2026-03-09 the open-source release of a cryptographic hardware/software stack for classical and post-quantum operations — the Cryptolib embedded library paired with a programmable Asymmetric Cryptography Coprocessor (ACC) with hardware-accelerated ML-KEM, ML-DSA, and SLH-DSA, reporting 6–9x speedups and 36–75% higher max operating frequency at near-zero area cost. ([businesswire.com — ZeroRISC and Leading Research Institutions Deliver Production Grade Post Quantum Cryptography for Open Silicon](#), as of 2026-03-09, confidence: medium, source: ingest)
- *No prior mention of ZeroRISC's PQC hardware/software stack in the knowledge base.*

OTA update signing (ota-update-signing)

New developments (2)

Firmware update signing schemes

- Microsoft plans to transition Windows Production signing to PQC-by-default in 2027, following CA replacement in 2026, with considerations for hybrid signatures and legacy systems. ([Preparing the Windows ecosystem for next-generation code signing](#), as of 2026-08-20, confidence: high, source: grok)
- *New vendor-specific migration timeline (Windows code signing) not present in existing knowledge.*

PQ migration paths

- IETF draft-ietf-jose-pq-composite-sigs describes PQ/T hybrid composite signatures for JOSE and COSE, combining ML-DSA with ECDSA or EdDSA for artifact and firmware signing. ([PQ/T Hybrid Composite Signatures for JOSE and COSE](#), as of 2026-08-21, confidence: high, source: grok)
- *Introduces a new composite-signature standardization effort (JOSE/COSE) distinct from the existing non-composite hybrid approach described in the pquip-hsm-constrained draft.*

Newly captured background -- older material, first time in the knowledge base (4)

Firmware update signing schemes

- UEFI Security Sub-Team published a whitepaper outlining PQC updates to the UEFI specification, including support for multiple signatures in PKCS#7 SignedData, TBSCertificate hash in DB, and updates to the UEFI CA for PQC algorithms like ML-DSA and SLH-DSA. ([POST-QUANTUM CRYPTOGRAPHY: UEFI SPECIFICATION UPDATES](#), as of 2026-04-29, confidence: high, source: grok)
- *New detail on UEFI specification-level PQC changes (multi-signature PKCS#7, CA updates) not previously covered in the knowledge base.*
- Western Digital adopted ML-DSA-87 with dual-signing (RSA-3072 hybrid) for secure boot, firmware, and package signing in enterprise HDDs, with PQC PKI/HSM deployed in 2025. ([Post-Quantum Cryptography for Enterprise HDD Security](#), as of 2026-05-15, confidence: medium, source: grok)
- *New vendor deployment example (Western Digital enterprise HDD firmware signing) not previously in the knowledge base.*

OTA update signing

- Android 17 integrates ML-DSA into Android Verified Boot (AVB) for quantum-resistant signatures in the boot sequence and updates Android Keystore and Google Play for ML-DSA app signing. ([Google Online Security Blog: March 2026](#), as of 2026-03-31, confidence: high, source: grok)

- *New platform-specific PQC secure-boot/signing adoption (Android AVB, Keystore, Play) not previously recorded.*
- SECQAI demonstrated the first PQC signing (FIPS 204 ML-DSA and FIPS 205 SLH-DSA) in a FOTA update process on its CHERI-based Q-Locked SE01 SoC with a global defence contractor. ([SECQAI and global defence contractor demonstrate world's first PQC in a FOTA process on a Memory Safe Chip](#), as of 2025-11-26, confidence: medium, source: grok)
- *New distinct hardware platform/vendor demonstration of PQC FOTA signing, separate from the existing ESP32-based demonstration.*

TPM & TCG specifications (tpm-and-tcg)

Newly captured background -- older material, first time in the knowledge base (2)

DICE

- DICE Attestation Architecture V1.3 RC1 and DICE Certificate Profile V1.2 RC1 entered public review with periods from July 23, 2026 to September 21, 2026. ([Public Review Specifications](#), as of 2026-07-23, confidence: high, source: grok)
- *Reports a new draft revision cycle (v1.3 RC1 / Certificate Profile v1.2 RC1) beyond the currently ratified v1.2, not previously recorded.*

TCG specs

- TCG published Platform Certificate Profile Version 2.1 on January 27, 2026, stating that the Root of Trust (RoT) is not limited to the TPM Endorsement Key and can be based on technologies such as DICE or MARS. ([TCG Platform Certificate Profile](#), as of 2026-01-27, confidence: high, source: grok)
- *New specification version broadening acceptable Root-of-Trust technologies beyond the TPM EK; not present in the base snapshot.*

Software supply chain

Platform code signing (platform-code-signing)

New developments (3)

C2PA content credentials — PQ signing planned, not shipped

- The C2PA specification's own explainer (current specification line 2.4) states: "Future support for ML-DSA algorithms (ML-DSA-44, ML-DSA-65, and ML-DSA-87) with approved parameters, as defined in NIST's FIPS 204, is planned for the C2PA standard." ([spec.c2pa.org — Explainer](#), as of 2026-08-31, confidence: medium, source: ingest)
- *No C2PA content-credentials coverage exists in the base snapshot; this establishes C2PA's PQC plans as a new subtopic.*
- The stated gate for adoption is library maturity, not standardization: ML-DSA will be added to the supported signature algorithm list "when stable ML-DSA support is broadly available in commonly used cryptographic libraries." No target date is given. ([spec.c2pa.org — Explainer](#), as of 2026-08-31, confidence: medium, source: ingest)
- *New detail on C2PA's adoption criteria for PQC, not previously captured anywhere in the base.*
- Current C2PA signing rests on standard X.509-based credentials with SHA2-256-class hashing in a Merkle-tree-like manifest structure; no post-quantum signature algorithm is in the currently supported set. ([spec.c2pa.org — Explainer](#), as of 2026-08-31, confidence: medium, source: ingest)

- *New baseline-state fact describing C2PA's current (non-PQC) cryptographic construction, absent from the existing knowledge base.*

Container signing & CI/CD (container-signing-cicd)

Superseded (1)

Container image signing

- Go 1.27.1 includes the crypto/mlDSA package implementing FIPS 204 ML-DSA in the standard library, with ML-DSA-44/65/87 parameter sets. ([mlDSA package - crypto/mlDSA - Go Packages](#), as of 2026-09-01, confidence: high, source: grok)
- supersedes: "[d0001:9] The Go standard library dependency Sigstore named as its gating condition is now nearly met: `crypto/mlDSA` implementing FIPS 204 is present in the Go standard library documentation at version go1.27rc2"
- *The gating condition (crypto/mlDSA in a stable Go release, not merely rc2) is now fully met with Go 1.27.1, a status change beyond a re-dated restatement.*

Artifact signing frameworks (artifact-signing-frameworks)

Newly captured background -- older material, first time in the knowledge base (3)

Sigstore/cosign

- As of March 2026, a community proof-of-concept demonstrates end-to-end ML-DSA-65 signing and verification in cosign (import-key-pair, sign-blob, verify-blob) using a patched Go 1.26.0-test build with crypto/mlDSA support. ([cosign with post-quantum MLDSA sample · sigstore/cosign · Discussion #4771](#), as of 2026-03-16, confidence: medium, source: grok, kind: chatter, status: unconfirmed, corroboration: single-source)
- *This is a distinct community demonstration using a patched Go stdlib build (crypto/mlDSA, import-key-pair) rather than the Trail of Bits custom Go/Rekor implementation already recorded, adding a genuinely new data point on cosign PQC feasibility.*
- Sigstore's algorithm registry currently lists ml-dsa-44, ml-dsa-65, and ml-dsa-87 as experimental pure variants that are not yet fully functional within Sigstore. ([architecture-docs/algorithm-registry.md at main · sigstore/architecture-docs](#), as of undated, confidence: medium, source: grok)
- *Provides new specificity on which ML-DSA parameter sets are registered and their current experimental/non-functional status, a detail not previously captured about the algorithm registry.*

TUF/Notary

- A November 2025 peer-reviewed paper details a phased PQC migration for TUF role hierarchies (root, targets, snapshot, timestamp) and Notary v2 signer certificates, starting with hybrids (RSA/EC + SPHINCS+/Dilithium/Falcon) then moving to pure PQC algorithms with specific signature size deltas. ([Quantum-Resilient PKI for Container Supply Chains: Post-Quantum Cryptography in TUF and Notary Frameworks](#), as of 2025-11-01, confidence: high, source: grok, kind: analysis)
- *Introduces a new, previously unrecorded phased migration proposal spanning both TUF and Notary v2 together, with concrete hybrid-then-pure algorithm staging not covered by existing TUF-only or Notary-only claims.*

PKI, certificates, identity & authentication

Public CA ecosystem (**public-ca-ecosystem**)

New developments (2)

CA/B Forum roadmap

- No new CA/Browser Forum ballot specifically enabling or requiring publicly-trusted PQC TLS server certificates had been adopted or merged into Baseline Requirements as of late August 2026; recent ballots (e.g., SMC017v2 adopted July 30, 2026) addressed RSA key sizes instead. ([Ballot SMC017v2: Increase Minimum RSA CA Key Size | CA/Browser Forum](#), as of 2026-08-27, confidence: medium, source: grok)
- *Introduces a genuinely new, checkable fact — the specific adopted ballot SMC017v2 on RSA CA key sizes as the Forum's most recent activity — not previously recorded, alongside continuing absence of a PQC TLS ballot.*

Public CAs

- DigiCert published updated recommendations for Post-Quantum TLS Migration on August 26, 2026, reiterating ML-DSA-only (not composite) as the default for PQC certificate authentication in public Web PKI contexts and noting that CA/Browser Forum discussions around post-quantum public trust certificates are focused on ML-DSA. ([Recommendations for Post-Quantum TLS Migration | DigiCert](#), as of 2026-08-26, confidence: high, source: grok)
- *Beyond reiterating the known ML-DSA-only default, this adds the new detail that ongoing CA/Browser Forum discussions on public-trust PQC certificates are specifically centered on ML-DSA, which was not previously recorded.*

Conflicting -- needs review (1)

CT logs

- Chrome Root Program updated its policy page on August 28, 2026, reaffirming that it has no immediate plan to add traditional X.509 certificates containing post-quantum cryptography to the Chrome Root Store and will pursue Merkle Tree Certificates (MTCs) as the primary path, with no planned implementation timelines for MTCs unless explicitly stated. ([Chrome Root Program - Moving Forward, Together](#), as of 2026-08-28, confidence: high, source: grok)
- conflicts with: "[d0002:10] Google's MTC rollout plan includes a Phase 2 targeted for Q1 2027 in which it will invite CT Log operators... to participate in the initial bootstrapping of public MTCs"
- *The August 28 Chrome Root Program policy statement that it has no planned implementation timelines for MTCs sits in tension with the earlier reported concrete Phase 2/Q1 2027 rollout timeline, and it's unclear whether the timeline was walked back or simply omitted here.*

Enterprise CA platforms (**enterprise-ca-platforms**)

New developments (1)

ADCS

- Microsoft Learn documentation updated 2026-08-25 details AD CS ML-DSA support phases, with Phase 1 (ML-DSA signatures) available and Phase 2 (ML-KEM, composite ML-DSA, composite ML-KEM) planned; requires Windows Server 2025 with KB5087539 or later on CA servers and Windows 11 24H2/25H2 with KB5067036 or later on clients. ([Post-Quantum Cryptography in AD CS overview - Windows Server](#), as of 2026-08-25, confidence: high, source: grok)
- *Adds new specifics — explicit Phase 2 scope (ML-KEM, composite ML-DSA, composite ML-KEM) and the*

previously unrecorded client-side KB5067036/Windows 11 24H2-25H2 requirement — not present in existing phased-rollout claims.

Superseded (1)

Venafi

- CyberArk Certificate Manager (Trust Protection Foundation) upgrade notes (Aug 2026) state that experimental PQ keys/certificates are invalidated upon upgrade to versions supporting final FIPS 204/205 standards; PQ algorithms are now available by default without a registry flag. ([Important Considerations before upgrading CyberArk Trust Protection Foundation](#), as of 2026-08-21, confidence: medium, source: grok)
- supersedes: "[d0002:19] Current Venafi documentation labels PQC signing key algorithms as 'Experimental': PQC environments must use software (not HSM) key storage... processing time is higher due to larger key/signature sizes, and PQC support is enabled by default on CodeSign Protect clients."
- *Updates the prior 'Experimental' status: PQ algorithms are now default-enabled without a registry flag and experimental-era PQ keys/certs are invalidated on upgrade to final-standard-supporting versions.*

Certificate formats & hybrid constructions (cert-formats-hybrid)

New developments (2)

RFC 3161 timestamping

- BeatQuantum launched a dedicated post-quantum RFC 3161 Time-Stamp Authority service issuing pure ML-DSA-87 or SLH-DSA-SHA2-256f signed timestamp tokens (no classical component) under its own root, effective 1 September 2026. ([Time-Stamp Authority | BeatQuantum](#), as of 2026-09-01, confidence: medium, source: grok)
- *New vendor product offering pure (non-hybrid) PQC-signed RFC 3161 tokens, not previously recorded among existing Ascertia/Codegic TSA product claims.*

X.509 composite and hybrid certs

- DigiCert recommends ML-DSA-only X.509 certificates as the default for PQC migration over composite ML-DSA certificates, citing broader industry convergence, implementation maturity in OpenSSL 3.5+, BoringSSL and Java 24+, and CA/B Forum focus on ML-DSA. ([Recommendations for Post-Quantum TLS Migration | DigiCert](#), as of 2026-09-02, confidence: medium, source: grok)
- *Adds a distinct vendor's (DigiCert) recommendation with new supporting detail (specific software maturity, CA/B Forum stance) beyond the existing EJBICA guidance already recorded, so it's new information rather than a mere restatement.*

Enterprise auth protocols (enterprise-auth-protocols)

Superseded (1)

LDAP/AD

- Microsoft Learn documentation for Post-Quantum Cryptography in AD CS (updated 2026-08-25) details phased rollout with ML-DSA in Phase 1 (available), ML-KEM and composite algorithms in Phase 2 (planned), and extends support plans to CEP, CES, NDES, and OCSP role services. ([Post-Quantum Cryptography in AD CS overview - Windows Server](#), as of 2026-08-25, confidence: high, source: grok)
- supersedes: "[d0004:2] Microsoft AD CS PQC documentation (updated 2026-08-07) specifies phased rollout: ML-DSA available in Phase 1, ML-KEM and composite algorithms planned for Phase 2; requires newly

installed CAs with no in-place migration of existing CAs."

- *Same phased-rollout structure as the 2026-08-07 version but adds a new detail (extension of support plans to CEP, CES, NDES, and OCSP role services), so it updates rather than merely reconfirms the prior claim.*

Newly captured background -- older material, first time in the knowledge base (1)

Kerberos

- A new individual IETF Internet-Draft draft-rsche-kitten-pkinit-crypto-deprec-00 (published 2026-06-23) specifies deprecation of RSA key transport, DH MODP group 2, SHA-1 KDF, and sha1WithRSAEncryption in Kerberos PKINIT, plus a new paChecksum2 field for checksum agility, updating RFCs 4556/5349/8636. ([Deprecation of Outdated Cryptographic Algorithms and Parameters in Kerberos PKINIT](#), as of 2026-06-23, confidence: high, source: grok)
- *This is a distinct IETF draft (different from draft-bokovoy-kitten-pkinit-pqc) addressing deprecation of legacy PKINIT crypto algorithms, not previously recorded.*

Passwordless & hardware tokens (passwordless-and-tokens)

New developments (1)

PIV/CAC smart cards

- As of August 27, 2026, GSA is updating the Federal Identity, Credential, and Access Management (FICAM) architecture to support post-quantum cryptography, and its laboratory for testing physical access control systems has begun evaluating quantum-resistant technology, with discovery/planning in 2026-2027, pilots in 2027-2028, and full migration targeted by 2035. ([US begins post-quantum upgrade of federal identity credentials | Biometric Update](#), as of 2026-08-27, confidence: high, source: grok)
- *Adds new detail on GSA's broader FICAM architecture update and physical access control lab testing with a distinct multi-phase timeline through 2035, beyond the previously known Shatterproof Digital Identity pilot.*

Revocation & status infrastructure (revocation-status-infra)

Superseded (1)

Merkle Tree Certificates

- Editor's copy of draft-ietf-plants-merkle-tree-certs (post-v05) as of 2026-08-28 includes renames (e.g., MerkleTreeCertEntry to MTCLogEntry), fixed test vectors, updates to CA certificate representation and subtree proofs, and clarifies CRLs/OCSPs apply unmodified to MTCs. ([Merkle Tree Certificates](#), as of 2026-08-28, confidence: high, source: grok)
- supersedes: "[d0004:5] Editor's copy of draft-ietf-plants-merkle-tree-certs (post-v05) includes renames (e.g., MerkleTreeCertEntry to MTCLogEntry), fixed test vectors, and updates to CA certificate representation and subtree proofs since July 2026."
- *Restates the same editor's-copy update with a fresher date but adds a genuinely new detail — explicit clarification that CRLs/OCSPs apply unmodified to MTCs — so it replaces rather than merely reconfirms the prior entry.*

Long-term signature validation & document signing (long-term-signature-validation)

New developments (1)

ETSI TS 119 312 algorithm catalogue transitions

- ETSI initiated work items for TR 119 330 series on PQC impact on electronic signatures and trust infrastructures (including signature creation/validation, TSPs, and EUDI Wallet), with adoption in June 2026 and early drafts targeted for 2027. ([Work Programme - Work Item Plan](#), as of 2026-08-29, confidence: high, source: grok)
- *New ETSI work-item information about a TR 119 330 series on PQC impacts across signatures/TSPs/EUDI Wallet, not previously recorded in the base or prior deltas.*

Identity provider platforms (idp-platforms)

Newly captured background -- older material, first time in the knowledge base (1)

Keycloak

- A new Keycloak GitHub issue (#50304, opened 2026-06-24, targeting milestone 27.0.0) requests support for PQ/T Hybrid Composite Signatures for JOSE and COSE to enable hybrid signatures combining ML-DSA with ECDSA or EdDSA during the transition to full PQC, referencing the active IETF draft-ietf-jose-pq-composite-sigs. ([Support PQ/T Hybrid Composite Signatures for JOSE and COSE · Issue #50304 · keycloak/keycloak · GitHub](#), as of 2026-06-24, confidence: high, source: grok)
- *This is a distinct, new tracked engineering item (hybrid composite signatures via a separate IETF draft) not previously recorded, complementing but not duplicating the existing ML-DSA-only OIDC signing issue (#50355) and the SAML PQC-readiness issue (#48825).*

Network & transport protocols

IPsec & VPN protocols (ipsec-and-vpn)

New developments (3)

IKEv2 with RFC 8784 and 9370

- draft-ietf-ipsecme-ikev2-prf-plus-02 (published August 28, 2026) specifies variable-length output PRFs for IKEv2 and explicitly applies to PQC extensions including RFC 8784, RFC 9370, and RFC 9867. ([Use of Variable-Length Output Pseudo-Random Functions \(PRFs\) in the Internet Key Exchange Protocol Version 2 \(IKEv2\)](#), as of 2026-08-28, confidence: high, source: grok)
- *New draft introducing variable-length PRF support tied to PQC IKEv2 extensions, not previously recorded.*
- Cisco IOS XE 26.2 adds native support for ML-DSA signatures (ML-DSA-44/65/87) for IKEv2 mutual authentication, enabling fully post-quantum IKEv2 tunnels when combined with ML-KEM-768 hybrid key exchange (previously available in 26.1). ([Post-Quantum Key Exchange on Cisco Routers - IPsec Series, Part 9](#), as of 2026-09-01, confidence: high, source: grok)
- *This is a new, distinct vendor implementation detail (Cisco IOS XE ML-DSA support for IKEv2 auth) not previously recorded; existing knowledge only covers strongSwan's experimental ML-DSA branch and Cisco's RFC 8784 PPK support.*

OpenVPN

- Easy-RSA added first-class PQC support (ML-DSA, ML-KEM, SLH-DSA, Falcon, hybrids) via PR #1432, with updates through August 2026, requiring OpenSSL 3.5+ or OQS provider. ([feat\(pqc\): add native Post-Quantum Cryptography support to Easy-RSA](#), as of 2026-08-15, confidence: high, source: grok)
- *New tooling development (Easy-RSA certificate generation) not previously covered in the OpenVPN section.*

Newly captured background -- older material, first time in the knowledge base (3)

IKEv2 with RFC 8784 and 9370

- Palo Alto Networks PAN-OS IKEv2 implementation via RFC 9242 and RFC 9370 supports a broad set of PQC KEMs including ML-KEM (512/768/1024), BIKE (L1/L3/L5), FrodoKEM variants, HQC (128/192/256), NTRU-Prime sntrup761, and Classic McEliece variants. ([How RFC 9242 and RFC 9370 Resist Quantum Computing Threats](#), as of 2026-07-20, confidence: high, source: grok)
- *Adds a specific, previously unstated list of supported PQC KEM algorithms beyond the general agility claim already recorded for PAN-OS.*
- draft-ietf-ipsecme-ikev2-downgrade-prevention reached the RFC Editor queue on July 8, 2026 after IESG review, defining IKE_SA_INIT_FULL_TRANSCRIPT_AUTH to prevent downgrade attacks in hybrid ML-KEM IPsec setups. ([IPsec Downgrade Attack Survives ML-KEM: Cloudflare Ships Authentication Fix](#), as of 2026-07-09, confidence: high, source: grok)
- *New draft and new mechanism (transcript authentication for downgrade prevention) not present in the knowledge base.*

WireGuard

- Surfshark launched post-quantum encryption on WireGuard for macOS, Linux, and Android in January 2026 using PSK-based hybrid handshake with NIST-approved Kyber. ([Can WireGuard Withstand the Quantum Storm?](#), as of 2026-04-26, confidence: medium, source: grok)
- *Adds a new VPN vendor (Surfshark) not previously mentioned among WireGuard PQC deployments.*

TLS 1.3 hybrid key exchange (tls-hybrid-keyex)

New developments (1)

X25519MLKEM768 adoption defaults

- As of August 28 2026, Google Cloud Load Balancing added X25519MLKEM768 support for Application and Proxy Load Balancers on a staged rollout (opt-in from June 2 2026, enabled by default from October 2026, permanently on with no opt-out from October 2027). ([ML-KEM vs X25519 TLS 1.3: 52% Adoption, +1.5ms \[2026\]](#), as of 2026-08-28, confidence: medium, source: grok)
- *New, specific rollout timeline detail for Google Cloud Load Balancing not previously covered (the base only mentions Google Cloud CDN generically).*

QUIC, HTTP/3, DTLS (quic-http3-dtls)

New developments (1)

QUIC

- The ant-quic Rust crate (v0.27.47, August 2026) implements pure post-quantum QUIC transport using only ML-KEM-768 for key exchange and ML-DSA-65 for authentication with no classical fallback. ([ant-quic - crates.io](#), as of 2026-08-24, confidence: high, source: grok)
- *New concrete implementation detail of a pure-PQC QUIC library not previously recorded.*

Newly captured background -- older material, first time in the knowledge base (2)

DTLS

- RFC 10024 (published August 2026) defines the hybrid groups X25519MLKEM768, SecP256r1MLKEM768,

and SecP384r1MLKEM1024 for TLS 1.3 and registers them with DTLS-OK: Y in the IANA TLS Supported Groups registry. ([RFC 10024 - Post-Quantum Traditional \(PQ/T\) Hybrid Key Agreement Mechanisms for TLS 1.3](#), as of 2026-08-01, confidence: high, source: grok)

- *This is a concrete, newly published RFC that formally registers specific hybrid groups with DTLS-OK: Y, going beyond the general framework guidance already known from RFC 9954.*

QUIC

- A December 2025 comparative study found that QUIC reduces the performance penalty (handshake latency and bandwidth overhead) of hybrid and pure post-quantum KEMs compared to TLS, especially under lossy network conditions. ([Comparative Analysis of Post-Quantum Handshake Performance in QUIC and TLS Protocols](#), as of 2025-12-01, confidence: high, source: grok)
- *New quantitative comparative finding about QUIC vs TLS PQ handshake performance not previously present in the knowledge base.*

Link & access layer (**link-access-layer**)

New developments (1)

802.1X/EAP

- The IAB is hosting a workshop on Accelerating the Deployment of Post-Quantum Authentication (pqws) in Prague on October 11-12, 2026, to gather deployment experience and address challenges with post-quantum signatures and authentication in protocols including EAP-TLS (noting large certificate chain issues per RFC 9191). ([IAB Workshop on Accelerating the Deployment of Post-Quantum Authentication \(pqws\)](#), as of 2026-08-31, confidence: high, source: grok)
- *New concrete event (IAB workshop, Prague, Oct 11-12 2026) addressing PQC deployment challenges in EAP-TLS not previously recorded.*

SSH (**ssh**)

Newly captured background -- older material, first time in the knowledge base (2)

OpenSSH

- OpenSSH 10.5 (released 2026-08-11) includes an upstream performance improvement for keying the sntrup761 key agreement algorithm. ([OpenSSH: Release Notes](#), as of 2026-08-11, confidence: high, source: grok)
- *New detail identifying the sntrup761 keying performance improvement as an upstream OpenSSH 10.5 change, distinct from the previously noted Debian 1:10.4p1 package-level improvement.*

mlkem768 keyex

- OpenSSH briefly implemented mlkem768brainpoolp256r1-sha256 (added July 2026) before removing it in August 2026 (commit 05ec77c14b9143b4d49f05a8dcd1366a92b9024e). ([OpenSSH: ChangeLog](#), as of 2026-08-08, confidence: high, source: grok)
- *Genuinely new fact about a short-lived key exchange algorithm addition and removal not present anywhere in the existing knowledge base.*

DNS & time (**dns-and-time**)

Newly captured background -- older material, first time in the knowledge base (2)

DNSSEC

- New IETF Internet-Draft draft-johani-dnsop-dnssec-alg-split-02 (published July 2026) proposes Algorithm-Split DNSSEC, separating large post-quantum KSK algorithms from small-signature ZSK algorithms to mitigate signature size issues in DNSSEC responses. ([draft-johani-dnsop-dnssec-alg-split-02 - Algorithm-Split DNSSEC: KSK/ZSK Algorithm Separation](#), as of 2026-07-03, confidence: high, source: grok)
- *Introduces a distinct new IETF proposal (KSK/ZSK algorithm split) not previously recorded, addressing signature size via a different mechanism than the existing dual-algorithm or MTL approaches.*
- New IETF Internet-Draft draft-kaizer-dnsop-ml-dsa-mtl-dnssec-01 (published August 2026) specifies ML-DSA-MTL (Module-Lattice-Based Signatures with Merkle Tree Ladders) for DNSSEC, applying MTL mode to ML-DSA-44 with SHAKE-128 to address large signature sizes. ([draft-kaizer-dnsop-ml-dsa-mtl-dnssec-01 - Module-Lattice-Based Signatures with Merkle Tree Ladders \(ML-DSA-MTL\) for DNSSEC](#), as of 2026-08-14, confidence: high, source: grok)
- *Provides a concrete new specification applying MTL mode to ML-DSA-44, a specific algorithm/draft combination not previously named (existing knowledge only mentioned MTL mode generically for SLH-DSA, Falcon, XMSS, LMS).*

HSMs, KMS & key management

QRNG & QKD adjacency (qrng-qkd)

New developments (3)

QKD developments

- SK Telecom secured Horizon Europe funding as the first Asian private company for a three-year multinational project to develop and demonstrate a next-generation QKD system based on Quantum Photonic Integrated Circuit-Artificial Intelligence (QPIC-AI), involving partners in Greece, Austria, Germany, and Korea. ([SK Telecom Develops Next-Generation Quantum Cryptography Technology with EU Research Funding](#), as of 2026-08-29, confidence: medium, source: grok)
- *New QKD R&D funding and consortium not previously recorded in the knowledge base.*

QRNG developments

- An experimental highly integrated broadband entropy source for QRNG based on vacuum fluctuations, using a hybrid laser-and-silicon-photonics chip, achieved a quantum random number generation rate of 67.9 Gbps under average conditional minimum entropy after digital equalization. ([Highly integrated broadband entropy source for quantum random number generators based on vacuum fluctuations](#), as of 2026-08-24, confidence: medium, source: grok)
- *New QRNG hardware result with a distinct throughput figure and chip design not covered by existing QRNG entries.*
- A tri-type QRNG system was demonstrated that on-demand generates uniform, Gaussian, and Rayleigh random numbers via quantum vacuum noise measurement, achieving over 42 Gbps secure uniform bit rate and over 14 Gbps secure Gaussian bit rate after extraction. ([40Gbps Tri-type Quantum Random Number Generator](#), as of 2026-08-24, confidence: medium, source: grok)
- *New QRNG capability (multi-distribution on-demand generation) not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

QKD developments

- Researchers demonstrated a practical side-channel-secure QKD (SCS-QKD) protocol over 200 km of fiber

with secure key rates of 196.03 bps at 200 km, 2.55 kbps at 150 km, and 18.31 kbps at 100 km using a stable 1.25 GHz system and finite-key analysis. ([Proof-of-principle experimental demonstration of side-channel-secure quantum key distribution over 200 km](#), as of 2026-08-11, confidence: high, source: grok)

- *New distinct QKD protocol demonstration and distance/key-rate results not present in existing QKD entries.*

Key escrow, archive & re-encryption (**key-escrow-archive-recovery**)

New developments (2)

key custody

- Thales launched the Luna 8 HSM with support for post-quantum cryptography algorithms, an upgradeable architecture for future standards, and targeting PQC readiness for enterprise and government key custody applications. ([Thales launches Luna 8 hardware security module in India, targeting post-quantum cryptography readiness for enterprises and government](#), as of 2026-08-29, confidence: medium, source: grok)
- *New Thales HSM product (Luna 8) with upgradeable PQC architecture is a distinct fact not covered by the existing Luna HSM firmware-update claim.*
- The Trusted Computing Group released guidance on August 24, 2026, to verify PQC-ready TPMs meeting PTP 1.07 requirements, defining 'TCG PQC-ready TPM' and 'TCG PQC-upgradable TPM' categories for hardware key custody during PQC transitions. ([New Guidance Helps Businesses Verify Quantum-Safe Hardware Claims](#), as of 2026-08-24, confidence: medium, source: grok)
- *Introduces a new TCG certification scheme and categories for TPM key custody not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

archive & re-encryption

- QuCloud proposes a hybrid framework combining E91 QKD, Kyber PKE, SHA-256 hashing, and custom proxy re-encryption for quantum-resistant cloud storage security and mitigation of harvest-now-decrypt-later risks on archived data. ([QuCloud : Enhancing cloud storage security by combining quantum key distribution, post-quantum cryptography, and custom proxy re-encryption](#), as of 2026-06-01, confidence: low, source: grok)
- *New academic proxy re-encryption/QKD hybrid proposal for archived cloud data, not previously recorded under archive & re-encryption.*

HSM vendors (**hsm-vendors**)

Conflicting -- needs review (1)

Utimaco

- Utimaco Quantum Protect algorithms (ML-KEM, ML-DSA, LMS) passed NIST CAVP validation with specific certificate references; SLH-DSA and additional schemes like XMSS supported via the package. ([Utimacos Nist Validated PQC Algorithms Setting Standard](#), as of 2026-07-31, confidence: medium, source: grok)
- conflicts with: "[d0005:3] Utimaco u.trust General Purpose HSM Se-Series ... supports Quantum Protect PQC extension with ML-KEM, ML-DSA, LMS, XMSS, and SLH-DSA in progress."
- *Claims SLH-DSA is already supported via the package, contradicting the established and more recently reconfirmed status that SLH-DSA remains in progress/on the roadmap.*

Newly captured background -- older material, first time in the knowledge base (2)

Entrust nShield

- Entrust nShield 5s HSM datasheets list specific ML-DSA signing performance (e.g., up to 2,780 TPS for ML-DSA-44 on high model) and support for 18 SLH-DSA variants plus LMS via Post-Quantum Option Pack. ([nShield 5s HSM datasheet](#), as of 2026-06-08, confidence: high, source: grok)
- *Introduces new granular performance and algorithm-variant-count details not previously recorded for nShield 5s.*

Thales

- Thales Luna 8 Network HSM is available now as a network appliance in three models (Luna 100, 150, 190) with a custom Thales cryptographic processor delivering up to 100x faster PQC algorithm performance and an upgradeable architecture for future standards. ([Introducing Luna 8: Our Fastest HSM Ever, Built for the Post-Quantum Era](#), as of 2026-08-04, confidence: high, source: grok)
- *Adds specific model names, current availability status, and a concrete 100x performance figure not present in the earlier general Luna 8 launch claim.*

Enterprise IoT & OT

GNSS & satellite links (gnss-satellite)

New developments (3)

satcom link encryptors

- WISeKey subsidiary WISeSat.Space signed a strategic cooperation agreement with Kaynes Space Technology (KSTPL) to manufacture post-quantum-secure WISeSat satellites in India, integrating SEALSQ post-quantum cryptographic semiconductors and WISeKey Root of Trust, establishing India as an official launch location alongside the US. ([WISeKey, Sealsq team with KSTPL to manufacture post-quantum satellites in India](#), as of 2026-08-27, confidence: medium, source: grok)
- *New manufacturing/partnership development for PQC-secured satellites not previously recorded, distinct from the existing SEALSQ QSOC launch claim.*
- On 2 September 2026, Cyprus and Greece established their first quantum-secure satellite data connection using Sparkle's Quantum Safe Interconnect (QSI) system and Hellas Sat's satellite infrastructure, protecting against harvest-now-decrypt-later risks. ([Cyprus and Greece Establish Quantum-Secure Satellite Link](#), as of 2026-09-02, confidence: medium, source: grok)
- *New satcom quantum-safe deployment event not present in the knowledge base.*
- On 1 September 2026, Arqit, Es'hailSat, and AIEE demonstrated quantum-safe encryption across live satellite-connected networks using Es'hail-1 and Es'hailSat's Doha teleport infrastructure with no adverse effect on performance. ([Arqit, Es'hailSat and AIEE Demonstrate Quantum-Safe Satellite Security](#), as of 2026-09-01, confidence: medium, source: grok)
- *New satcom quantum-safe demonstration event not present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (4)

Galileo OSNMA and GNSS signal authentication

- Galileo SAS initial capability started permanently in December 2025 with L3 satellites (E14 and E18) having E6-C encryption; full constellation E6-C encryption is planned for the second half of 2026, with SAS Initial Service Declaration foreseen for early 2027. ([Beyond OSNMA: Signal Authentication Service](#), as of 2026-02-27, confidence: high, source: grok)
- *Provides specific satellite identifiers, rollout timeline, and an ISD target date not previously recorded.*
- Publication of the Galileo SAS ICD and Guidelines is planned for 2026. ([Beyond OSNMA: Signal Authentication Service](#), as of 2026-02-27, confidence: high, source: grok)

- *New detail about SAS documentation timeline not present in the knowledge base.*
- As of July 2026, Oscilloquartz has integrated Galileo OSNMA authentication into its OSA5412, OSA5422, and OSA5430/40 timing and synchronisation platforms using u-blox GNSS technology. ([Galileo OSNMA enhances secure GNSS timing for critical infrastructure](#), as of 2026-07-30, confidence: high, source: grok)
- *New commercial product integration of OSNMA not previously recorded.*

satcom link encryptors

- Voyager Technologies and IBM demonstrated the first post-quantum-secured communication link between the International Space Station and Earth in April 2026, using IBM Quantum Safe Remediator (with NIST PQC standards ML-KEM/FIPS 203, ML-DSA/FIPS 204, SLH-DSA/FIPS 205) deployed on Voyager's Space Edge micro datacenter as a proxy for legacy applications without code changes. ([Voyager and IBM Prove Post-Quantum Security On-Orbit](#), as of 2026-04-15, confidence: high, source: grok)
- *New, corroborated demonstration of a PQC-secured ISS-Earth link, distinct from the previously recorded QuSecure Starlink demonstration.*

Payments hardware (payments–hardware)

New developments (1)

EMV

- IDEMIA Secure Transactions was awarded the top ranking in Juniper Research's Post-Quantum Cryptography Competitor Leaderboard for its end-to-end PQC solutions including hardware accelerators and smart card implementations. ([Idemia receives top listing in post-quantum security](#), as of 2026-09-02, confidence: high, source: grok)
- *New third-party competitive ranking of IDEMIA's PQC portfolio not present in the existing knowledge base.*

Superseded (1)

EMV

- STMicroelectronics ST54M secure mobile chip with PQC hardware accelerator for ML-KEM and ML-DSA, integrating NFC/eSE/eSIM, has completed Common Criteria 2022 EUCC and EMVCo certification testing with production and full certification targeted for July 2026. ([STMicroelectronics unveils world's first ST54M secure mobile chip with post-quantum cryptography for next-generation connected services](#), as of 2026-06-24, confidence: high, source: grok)
- supersedes: "[d0004:2] STMicroelectronics ST54M secure mobile chip integrates NFC, eSE, eSIM and a KECCAK hardware accelerator for ML-KEM and ML-DSA, targeting EMVCo certification and applications including contactless payments."
- *Adds concrete certification progress (CC EUCC completed, EMVCo testing completed) and a specific July 2026 target date beyond the prior claim's vague 'targeting EMVCo certification'.*

Automotive (automotive)

New developments (2)

V2X

- PQAKA, a post-quantum authentication and key agreement protocol for 5G-V2X using ML-KEM, achieves mutual authentication and forward secrecy while maintaining compatibility with existing 3GPP entities and deployability on USIM/HSM/TEE platforms. ([Post-Quantum Authentication and Key Agreement Protocol for](#)

[5G-V2X Communication](#), as of 2026-08-19, confidence: medium, source: grok)

- *Introduces a new named protocol (PQAKA) and technical approach for 5G-V2X PQC authentication not present in the existing knowledge base.*
- A 2026-08-24 arXiv study on C-V2X sidelink found Falcon-512 the only NIST PQC signature that fits SAE J3161/IEEE 1609.2 transport-block constraints, achieving mean latency ~52 ms but meeting 90% PDR threshold only at low traffic levels under LOS; Dilithium-2 and SPHINCS+ exceeded capacity. ([Deployment Feasibility Analysis of Post-Quantum Digital Signatures in Safety-Critical C-V2X Communication for Urban Mobility Scenario](#), as of 2026-08-24, confidence: high, source: grok)
- *Provides a new, specific empirical finding on which PQC signature algorithm actually fits V2X transport-block/latency constraints, complementing but not restating the prior general size-mismatch claim.*

Newly captured background -- older material, first time in the knowledge base (2)

UNECE R155/156

- Using deprecated classical asymmetric algorithms may risk failing Type Approval under UNECE R155. ([On-demand webinar: Adoption of post-quantum cryptography in real-time automotive systems](#), as of 2026-03-10, confidence: medium, source: grok)
- *Adds a specific new risk linkage (algorithm deprecation to type-approval failure) not stated in existing R155 claims about CSMS/SUMS evidence requirements.*

V2X

- Post-quantum cryptography for intelligent transportation systems: An implementation-focused review (Vehicular Communications, Volume 60, August 2026) identifies thirteen research gaps in PQC adoption for in-vehicle networks and V2X, including non-PQC-ready standards and certificate-management barriers. ([Post-quantum cryptography for intelligent transportation systems: An implementation-focused review](#), as of 2026-08-01, confidence: high, source: grok)
- *Same source article as [d0001:7] but surfaces a distinct, previously unrecorded finding (thirteen research gaps in PQC adoption for ITS/V2X) beyond the AUTOSAR crypto-stack detail already captured.*

Industrial control protocols (industrial-protocols)

Conflicting -- needs review (1)

BACnet

- Veridify's DOME product provides device-level zero-trust security with support for post-quantum algorithms (ML-KEM/FIPS 203, ML-DSA/FIPS 204, and FALCON) for BACnet/SC, BACnet/IP, Modbus TCP, DNP3, OPC UA, and other industrial protocols, enabling retrofit on legacy devices without network changes. ([DOME - Veridify Security](#), as of 2025-10-20, confidence: medium, source: grok)
- conflicts with: "[d0001:2] no vendor-shipped concrete PQC integration for OPC UA was found"
- *This is an unconfirmed vendor self-report of a shipping product with PQC support across OPC UA, Modbus, DNP3, and BACnet, which contradicts prior survey findings of no shipped vendor implementation, so it is flagged rather than silently accepted or used to supersede.*

Newly captured background -- older material, first time in the knowledge base (1)

SCADA/PLC vendors

- A 2026 IEEE paper introduces a hybrid classical-post-quantum migration framework specifically for brownfield SCADA environments, including a migration taxonomy distinguishing coexistence, cryptographic hybridization, dual-root trust, protected conduits, and replacement-driven uplift, plus a conduit-first sequence prioritizing

gateways and management planes. ([Hybrid Classical–Post-Quantum Migration Framework for Brownfield SCADA Environments](#), as of 2026-05-16, confidence: high, source: grok)

- *No prior knowledge exists under the SCADA/PLC vendors subtopic, and this introduces a substantively new migration taxonomy and sequencing framework for brownfield SCADA PQC migration.*

Energy & grid (energy-grid)

New developments (1)

Grid infrastructure

- In power sector OT environments, migration of the IT-heavy portion (internet-facing systems, PKI, VPNs) to PQC is realistic by around 2030, while OT gateway, communication infrastructure, and field device protocols (RTU/IED level) face a probable additional 3-4 year delay beyond 2030-2031 due to legacy protocols and real-time constraints. ([Quantum Readiness for OT Is More Than Encryption](#), as of 2026-09-02, confidence: medium, source: grok)
- *Provides a new, more granular technical timeline distinguishing IT vs OT/field-device PQC migration pace in power grids, not previously captured in the base's regulatory-deadline-focused claims.*

Physical security devices (physical-security-devices)

New developments (1)

access control

- GSA's Physical Access Control System (PACS) laboratory is expanding its infrastructure to test quantum-resistant technology for federal building access systems and employee identification cards as part of FICAM modernization for post-quantum cryptography. ([US begins post-quantum upgrade of federal identity credentials](#), as of 2026-08-27, confidence: medium, source: grok)
- *New fact about GSA PACS lab testing PQC for federal access control/ID cards, not present in base snapshot.*

Newly captured background -- older material, first time in the knowledge base (3)

IP cameras

- A post-quantum security architecture for AI-driven surveillance systems integrates PQC mechanisms at session establishment and authentication levels while operating with existing IP networks, TLS channels, and VMS/NVR ecosystems without altering conventional symmetric encryption. ([A post-quantum security architecture for AI-driven surveillance systems: ensuring data protection in the quantum computing era](#), as of 2026-02-18, confidence: medium, source: grok)
- *Introduces a new academic architecture proposal for PQC integration in NVR/VMS surveillance systems not covered by existing base claims.*
- Eagle Eye Security Solutions advises selecting modern enterprise-grade IP camera and NVR hardware with upgradable crypto agility to support NIST-approved PQC algorithms via firmware updates. ([Preparing for Q-Day: Quantum-Resistant Security Cameras in Canada](#), as of 2026-05-12, confidence: low, source: grok)
- *New vendor guidance on crypto-agile hardware procurement distinct from existing Axis design claim.*

access control

- Secure intercom software playbook states that door devices with 10-year lifespans should begin shipping firmware with NIST-standardized PQC algorithms (ML-KEM, ML-DSA) in 2026 to address harvest-now-decrypt-later threats, driven by EU Cyber Resilience Act requirements. ([Secure Intercom Software: 2026 Buyer](#)

[& Builder Playbook](#), as of 2026-08-10, confidence: medium, source: grok)

- *New specific guidance on intercom/door device firmware timelines and regulatory driver not previously captured.*

Medical devices (medical-devices)

Superseded (1)

Medical devices

- A July 2026 Frontiers in Health Services review article on post-quantum cryptography for healthcare recommends a staged hybrid crypto-agile migration strategy for connected medical devices, including prioritizing long-confidentiality data, deploying hybrid key establishment at gateways, and requiring vendor update pathways and cryptographic dependency information in premarket submissions to align with FDA expectations under FD&C Act Section 524B. ([Post-quantum cryptography for healthcare: securing medical data, connected devices, and digital health infrastructure](#), as of 2026-07-16, confidence: medium, source: grok)
- supersedes: "[d0001:7] recommends a staged migration roadmap to minimize patient-safety disruption while reducing harvest-now-decrypt-later exposure"
- *Same Frontiers review, now dated and far more specific about the staged migration strategy (hybrid gateways, vendor update pathways, Section 524B linkage) than the vague 'staged migration roadmap' previously recorded.*

Newly captured background -- older material, first time in the knowledge base (1)

Regulatory guidance

- The July 2026 Frontiers review notes that FDA medical-device cybersecurity guidance emphasizes secure design, updates, and lifecycle risk management but does not currently require a formal cryptographic bill of materials (CBOM). ([Post-quantum cryptography for healthcare: securing medical data, connected devices, and digital health infrastructure](#), as of 2026-07-16, confidence: medium, source: grok)
- *New specific detail (absence of a CBOM requirement) not previously present in the knowledge base.*

Short-range IoT (short-range-iot)

New developments (1)

BLE

- NIST-standardized ML-KEM, ML-DSA, and SLH-DSA parameter sets all execute successfully on ESP32-C6 microcontroller without external memory, with measurements of latency, key/signature sizes, and memory use under embedded constraints including BLE telemetry scenarios. ([Exploring hardware implementation feasibility of post-quantum cryptography in embedded systems: Evaluation of NIST-standardized ML-KEM, ML-DSA and SLH-DSA on ESP32-C6](#), as of 2026-08-18, confidence: medium, source: grok)
- *New empirical evidence that all three ratified NIST PQC algorithms run successfully on a specific low-power microcontroller in BLE telemetry contexts, not previously recorded in the knowledge base.*

Developer & migration tooling

Interop & test tooling (interop-test-tooling)

New developments (3)

Interop test servers

- A new free online tool at <https://www.mylesagnew.com/pqctest/> performs a real post-quantum TLS 1.3 handshake against any public HTTPS server on port 443 to test support specifically for the hybrid group X25519MLKEM768 (ML-KEM-768 per FIPS 203). ([Post-Quantum Server Test | Myles Agnew](#), as of 2026-08-25, confidence: high, source: grok)
- *A newly identified interop test tool distinct from existing entries (OQS server, pqcscan), not previously recorded.*
- Cloudflare Radar provides a publicly accessible website post-quantum TLS key exchange support test widget that performs a TLS handshake to check support for hybrid post-quantum key exchange methods such as X25519MLKEM768 on port 443 (or other specified ports). ([Cloudflare Radar](#), as of 2026-09-03, confidence: high, source: grok)
- *Cloudflare Radar's PQ test widget is a distinct interop test tool not previously recorded, alongside the mylesagnew.com tool and pqcscan already in the base.*

tls-scan

- The openshift/tls-scanner Go module (for OpenShift/Kubernetes TLS compliance scanning) incorporates post-quantum (ML-KEM) readiness checks via testssl.sh, with a dedicated -pqc-check flag that exits non-zero on failure to detect TLS 1.3 + ML-KEM support. ([tls-scanner module - github.com/openshift/tls-scanner](#), as of 2026-08-25, confidence: high, source: grok)
- *New tooling detail about a distinct tls-scan-related project integrating PQC checks via testssl.sh, not previously present in the knowledge base.*

Crypto discovery & CBOM (crypto-discovery-cbom)

Superseded (1)

SBOM integration

- CISA published the final 2026 Minimum Elements for SBOMs on July 29, 2026, adding requirements for SBOM author digital signature, component license metadata, and restricting machine-processable formats to SPDX and CycloneDX only (dropping SWID). ([CISA's 2026 SBOM Minimum Elements Now Require Post-Build Data - OPSWAT](#), as of 2026-07-29, confidence: high, source: grok)
- supersedes: "[d0002:5] CISA's 2026 SBOM Minimum Elements can be satisfied by CycloneDX 1.6 (with exact property paths mapped for each field)"
- *Confirms finalization date and adds concrete new requirements (digital signature, license metadata, dropping SWID) beyond the prior mapping-focused claim.*

Newly captured background -- older material, first time in the knowledge base (2)

Crypto discovery & CBOM

- sbomify v26.7.0 (July 2026) added native CBOM support with NIST-grounded PQC readiness classification and post-quantum posture cards for components. ([Announcing sbomify v26.7.0: The One That Gets Quantum-Ready | Sbmify](#), as of 2026-07-07, confidence: medium, source: grok)
- *Identifies a new tool with CBOM/PQC readiness features not previously recorded.*

SBOM integration

- CISA's finalized 2026 SBOM Minimum Elements emphasize post-build data such as executable artifact

hashes and require scans of binaries/container layers rather than source dependencies alone. ([CISA's 2026 SBOM Minimum Elements Now Require Post-Build Data - OPSWAT](#), as of 2026-07-29, confidence: high, source: grok)

- *Introduces a new scope requirement (post-build/binary artifact data) not previously captured in the knowledge base.*

Performance & side-channel tooling (**perf-sidechannel**)

Superseded (1)

constant-time verification

- LFX mentorship report details extension of constant-time tooling in liboqs using Valgrind-Varlat and MemSan, successfully identifying a timing leak in NTRU Prime (sntrup) via nonzero mask function. ([LFX Mentorship – Enhancing Constant-Time Analysis Tooling in liboqs](#), as of 2026-09-03, confidence: medium, source: grok)
- supersedes: "[d0003:1] An LFX mentorship project completed February 2026 evaluated and extended constant-time analysis tools for liboqs, with plans to integrate them into the project's CI pipeline"
- *Same LFX mentorship project now specified with concrete tools used (Valgrind-Varlat, MemSan) and a specific finding (NTRU Prime timing leak), adding substantive detail beyond the prior general statement.*

Newly captured background -- older material, first time in the knowledge base (3)

constant-time verification

- liboqs 0.16.0 release (July 2026) removed the legacy HQC constant-time test as part of updates to algorithm support and testing infrastructure. ([liboqs/RELEASE.md](#), as of 2026-07-09, confidence: high, source: grok)
- *New, specific fact about a liboqs release change to constant-time testing infrastructure not covered by the existing LFX mentorship claim.*
- New toolchain automates TIMECOP, Binsec/Rel2, duedect, and RTLf for systematic constant-time and timing-leakage analysis of NIST PQDSS round-1/round-2 implementations, identifying overlapping and distinct findings across tools. ([Systematic Timing Leakage Analysis of NIST PQDSS Candidates: Tooling and Lessons Learned](#), as of 2026-08-11, confidence: medium, source: grok)
- *Introduces a new multi-tool automated constant-time verification toolchain not previously recorded in the knowledge base.*

fuzzing

- A May 2026 master's thesis from University of Oulu extends Cryptofuzz with PQC-specific support for differential fuzzing of ML-KEM and ML-DSA implementations across libraries including Botan, OpenSSL, and liboqs, executing ~106M ML-KEM and ~26M ML-DSA test cases. ([A DIFFERENTIAL FUZZING FRAMEWORK FOR ML-KEM AND ML-DSA IMPLEMENTATIONS IN CRYPTOGRAPHIC LIBRARIES](#), as of 2026-05-01, confidence: medium, source: grok)
- *New fuzzing tooling development for ML-KEM/ML-DSA not previously recorded (existing fuzzing entry only covers HQC differential fuzzing by Quarkslab).*

Crypto libraries & providers

PKCS#11 and provider plumbing (**pkcs11-plumbing**)

New developments (1)

PKCS#11 standard

- No PKCS#11 v3.3 standard, errata addressing external-mu ML-DSA signing, or related updates published as of 2026-09-03. ([PKCS #11 Specification Version 3.2 \(OASIS Standard\)](#)), as of 2026-09-03, confidence: medium, source: grok)
- *A checkable state-of-the-world fact (no v3.3/errata exists yet) not previously recorded, distinct from the earlier v3.2-lacks-external-mu claim.*

Superseded (1)

wolfPKCS11

- wolfPKCS11 release 2.1 (2026-06-25) added post-quantum cryptography support including ML-DSA and ML-KEM, plus CMake build support. ([GitHub - wolfSSL/wolfPKCS11: wolfSSL PKCS#11](#)), as of 2026-06-25, confidence: medium, source: grok)
- supersedes: "[d0003:2] wolfPKCS11 added support for ML-KEM and ML-DSA (features of PKCS#11 v3.2), configurable via --enable-milkem and related flags in wolfSSL"
- *Pins the earlier general claim to a specific release (2.1, 2026-06-25) and adds the new detail that this release also introduced CMake build support.*

Newly captured background -- older material, first time in the knowledge base (2)

libp11 provider

- OpenSC/libp11 version 0.4.20 (released 2026-08-08) added PKCS#11 3.2 interface discovery, updated bundled header to v3.2, and PKCS#11 provider support for ML-KEM-512/768/1024 key generation, encapsulation and decapsulation plus the PKCS11_evp_pkey_decapsulate() API. ([Releases · OpenSC/libp11 · GitHub](#)), as of 2026-08-08, confidence: high, source: grok)
- *libp11's PKCS#11 3.2 discovery support and ML-KEM key-management/encapsulation API are not covered anywhere in the existing base.*
- OpenSC/libp11 version 0.4.19 (released 2026-07-21) added native KEYMGMT, SIGNATURE and ASYM_CIPHER operations to the PKCS#11 provider and support for ML-DSA, SLH-DSA and FALCON key generation, signing and verification. ([Releases · OpenSC/libp11 · GitHub](#)), as of 2026-07-21, confidence: high, source: grok)
- *New, distinct piece of information about libp11's OpenSSL 3 provider operation support for ML-DSA/SLH-DSA/FALCON, not previously in the base.*

OpenSSL and forks (openssl-and-forks)

Conflicting -- needs review (1)

BoringSSL

- BoringSSL added ML-DSA signature support (ML-DSA-44/65/87) around April 2026. ([Cryptography | Noise](#)), as of 2026-04-01, confidence: medium, source: grok)
- conflicts with: "[d0002:1] BoringSSL added ML-DSA support including ML-DSA-87 for X.509 certificate validation and OIDs/NIDs for ML-DSA and ML-KEM via commits in October 2025"
- *This claim dates BoringSSL's ML-DSA addition to around April 2026, contradicting the established October 2025 commit dates without clearly superseding them.*

Newly captured background -- older material, first time in the knowledge base (1)

s2n-tls

- s2n-tls's default_pq policy that includes ML-DSA support is specifically version 20250721. ([Post Quantum Support](#), as of 2026-06-15, confidence: high, source: grok)
- *Adds a previously unrecorded specific policy version number to the existing general claim about s2n-tls ML-DSA/default_pq support.*

Language-native crypto stacks (language-native-crypto)

Newly captured background -- older material, first time in the knowledge base (2)

.NET CNG

- In .NET 10, MLDSA and SlhDsa members were renamed from *SecretKey* to *PrivateKey* (e.g., `ImportMLDsaPrivateKey`, `PrivateKeySizeInBytes`) as a breaking change for alignment with other asymmetric types. ([Breaking change - MLDSA and SlhDsa 'SecretKey' members renamed](#), as of 2026-07-31, confidence: high, source: grok)
- *New API-naming detail about .NET 10 PQC types not previously captured in the knowledge base.*

Java JCA + BouncyCastle

- Bouncy Castle Java 1.85 (July 28, 2026) adds support for hybrid certificates, post-quantum keys and signatures in X.509, and standards-based PKI migration mechanisms. ([New Release: Bouncy Castle Java 1.85](#), as of 2026-07-28, confidence: medium, source: grok)
- *Specifies concrete new PKI/X.509 capabilities in 1.85 beyond the vague 'significant advances' summary already recorded.*

Embedded & alternative libraries (embedded-alt-libraries)

Newly captured background -- older material, first time in the knowledge base (5)

GnuTLS

- GnuTLS 3.8.13 (released April 29, 2026) added support for deriving an ML-DSA public key from an expanded private key per RFC 9881. ([Blame · NEWS · master · gnutls / GnuTLS · GitLab](#), as of 2026-04-29, confidence: medium, source: grok)
- *Base snapshot only notes 3.8.13's release date via Wikipedia; this adds a specific new feature (RFC 9881 ML-DSA public-key derivation) not previously recorded.*

wolfSSL

- wolfSSL 5.9.2 (released June 30, 2026) renamed its post-quantum signature implementation from Dilithium to ML-DSA, updated SLH-DSA sign/verify APIs to require pre-hashed message digests, added RFC 9802 LMSS/XMSS support in X.509 certificate and CSR generation, and added ML-KEM support for PKCS#11. ([wolfSSL 5.9.2 release blog - wolfSSL](#), as of 2026-06-30, confidence: medium, source: grok)
- *New release-specific details (API renames, RFC 9802 cert/CSR support, PKCS#11 ML-KEM) not present in the base snapshot.*
- wolfPKCS11 2.1.0 (released July 2, 2026) added post-quantum crypto support for ML-DSA (FIPS 204) and ML-KEM (FIPS 203). ([wolfPKCS11 2.1.0 is out! \(X post\)](#), as of 2026-07-02, confidence: medium, source: grok)
- *Genuinely new claim about a specific wolfPKCS11 release adding ML-DSA/ML-KEM, not previously recorded (blocked retrieval, so added rather than used to supersede anything).*
- wolfSSL added PQC support (ML-KEM, ML-DSA, SLH-DSA, LMS, XMSS) to its Java providers wolfJSSE and wolfJCE, including PQC-enabled TLS 1.3 hybrid key exchange. ([Java is now #PostQuantum ready with wolfSSL! \(X post\)](#), as of 2026-08-11, confidence: medium, source: grok)
- *New information about PQC reaching wolfSSL's Java providers, not covered by any existing base claim.*

- wolfSSL enabled post-quantum and hybrid TLS 1.3 key exchange (including multiple SecPMLKEM and XMLKEM groups) when using curl built with wolfSSL. ([Post-Quantum Cryptography in curl - wolfSSL](#), as of 2026-06-16, confidence: medium, source: grok)
- *New integration detail (curl + wolfSSL PQC hybrid groups) absent from the base snapshot.*

Application dev stacks & languages

Language runtime TLS surfaces (`language-runtimes-tls`)

Superseded (2)

C/C++

- wolfProvider (July 2026) adds CAVP-validated ML-KEM and ML-DSA (plus hybrids) to OpenSSL 3.x applications via wolfCrypt backend, with no application code changes required. ([wolfProvider Post-Quantum Cryptography: ML-KEM and ML-DSA - wolfSSL](#), as of 2026-07-02, confidence: high, source: grok)
- supersedes: "[d0002:6] wolfProvider (July 2026) adds CAVP-validated ML-KEM/ML-DSA support to OpenSSL 3.x applications via wolfCrypt backend"
- *Same wolfProvider claim now specifies additional detail (hybrid group support and no required application code changes) beyond the original.*

Python

- pyca/cryptography >=48 (released June 2026) adds native support for ML-KEM (FIPS 203) and ML-DSA (FIPS 204) via Rust bindings and AWS-LC backend, making PQC one pip install away. ([Shipping post-quantum cryptography to Python - The Trail of Bits Blog](#), as of 2026-06-30, confidence: high, source: grok)
- supersedes: "[b1:22] pyca/cryptography has an open tracking/roadmap issue (as of April 2026) discussing whether and how to add ML-KEM, ML-DSA, and SLH-DSA support, indicating these algorithms are not yet part of its core API"
- *The roadmap discussion has resolved into shipped native support as of pyca/cryptography 48 (June 2026), superseding the earlier 'not yet part of core API' status.*

Newly captured background -- older material, first time in the knowledge base (2)

C/C++

- wolfSSL 5.9.2 (June 2026) adds ML-DSA (renamed from Dilithium), SLH-DSA, LMS/XMSS support, PKCS11 for ML-KEM, and removes liboqs integrations in favor of native implementations. ([wolfSSL 5.9.2 release blog - wolfSSL](#), as of 2026-06-30, confidence: high, source: grok)
- *New fact about core wolfSSL C library shipping native ML-DSA/SLH-DSA/LMS-XMSS and dropping liboqs, distinct from the earlier Rust-API roadmap claim and not previously recorded in the C/C++ subtopic.*

Go

- Go 1.27 adds a new crypto/mldsa package implementing ML-DSA (FIPS 204) with three parameter sets, plus support in crypto/x509 and crypto/tls for TLS 1.3 signatures. ([Go 1.27 brings generic methods, post-quantum crypto and a new JSON engine - Northeast Times](#), as of 2026-08-02, confidence: medium, source: grok)
- *First mention of ML-DSA signature support (crypto/mldsa, x509/tls integration) in Go, distinct from the existing ML-KEM key-exchange claims.*

Database & messaging drivers (`data-layer-drivers`)

New developments (1)

Postgres

- PostgreSQL 18 renames `ssl_ecdh_curve` to `ssl_groups`, changes its default to include X25519, and supports configuring TLS 1.3 groups including the hybrid PQC group X25519MLKEM768 (when linked to OpenSSL 3.5+); pre-18 versions cannot easily use such hybrid schemes. ([PostgreSQL 18 Release Notes](#), as of 2026-09-01, confidence: high, source: grok)
- *New server-side configuration detail (`ssl_ecdh_curve` renamed to `ssl_groups`, new default, PG18-specific hybrid group support) not previously present in the knowledge base.*

Cloud infrastructure

CDN & edge (`cdn-edge`)

New developments (2)

Cloudflare

- As of August 28, 2026, Cloudflare's PQC in Cloudflare products documentation confirms key agreement is enabled with X25519MLKEM768 for both visitor-to-Cloudflare and Cloudflare-to-origin connections, signatures are enabled with ML-DSA for Cloudflare-to-origin via AOP/COTS but only planned via Merkle Tree Certificates for visitor-to-Cloudflare connections, and internal network key agreement is mostly migrated while internal signatures are not yet implemented. ([PQC in Cloudflare products · Cloudflare SSL/TLS docs](#), as of 2026-08-28, confidence: high, source: grok)
- *Adds the previously unrecorded detail that visitor-to-Cloudflare signature authentication is planned via Merkle Tree Certificates, beyond what prior claims about origin authentication and internal migration status covered.*

Fastly

- Fastly, together with Cloudflare, accounts for nearly 70% of all observed PQ-TLS deployments on the internet as of measurement rounds through March 2026. ([Mind the Gap: Policy vs Reality in Post-Quantum TLS Deployment](#), as of 2026-08-24, confidence: medium, source: grok)
- *Introduces a new market-share statistic on PQ-TLS deployment share not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

Akamai

- As of August 13, 2026, Akamai made its Post Quantum Cryptography (PQC) to Origin and PQC Client to Edge behaviors available in limited availability for Standard TLS hostnames (in addition to Enhanced TLS). ([Aug 13, 2026 - Post Quantum Cryptography support for Standard TLS](#), as of 2026-08-13, confidence: high, source: grok)
- *Extends Akamai's PQC availability beyond Enhanced TLS (established in [d0003:1]) to Standard TLS hostnames, a genuinely new scope detail not previously recorded.*

SASE & zero trust (`sase-zero-trust`)

Conflicting -- needs review (2)

Netskope

- Netskope natively integrated NIST-approved post-quantum cryptography algorithms into its SASE platform across more than 120 data centers globally in Q2, delivering quantum-resilient encryption for customer transitions to quantum-safe environments. ([Transcript: Netskope Q2 2027 Earnings Conference Call](#), as of 2026-09-02, confidence: high, source: grok)
- conflicts with: "[d0002:1] Netskope NG-SWG Release 137 introduces beta support for the X25519MLKEM768 hybrid PQC key-exchange algorithm"
- *Earnings-call claim of native, global (120+ data center) integration contradicts the documented beta/feature-flag-gated status without clarifying whether this is a status upgrade or promotional overstatement.*
- Netskope One Next Gen SWG now supports X25519MLKEM768 hybrid PQC key exchange in commercial environments (client-to-proxy and proxy-to-server), implemented natively via OpenSSL 3.5 upgrade and available via tenant-level feature flag. ([Post-quantum cryptography \(PQC\) just landed in Netskope One Next Gen SWG](#), as of 2026-08-25, confidence: medium, source: grok)
- conflicts with: "[d0002:1] Netskope NG-SWG Release 137 introduces beta support for the X25519MLKEM768 hybrid PQC key-exchange algorithm"
- *Blocked-source claim describes the same feature as available in 'commercial environments' with a native OpenSSL 3.5 implementation, an apparent status update beyond documented beta that cannot be used to supersede on its own.*

Newly captured background -- older material, first time in the knowledge base (1)

Zscaler

- Zscaler Internet Access provides real-time SSL/TLS decryption, deep content inspection, and policy enforcement on traffic using hybrid or pure PQC key-encapsulation mechanisms (including ML-KEM), with auto-detection of ML-KEM and Open Quantum Safe groups alongside classical curves. ([Post-Quantum Cryptography \(PQC\) Inline Inspection with Zscaler Internet Access](#), as of 2026-07-17, confidence: medium, source: grok)
- *Adds new technical detail beyond the existing hybrid-only description: support for pure (non-hybrid) PQC KEMs and auto-detection of Open Quantum Safe groups.*

Confidential computing attestation (confidential-computing)

Newly captured background -- older material, first time in the knowledge base (1)

TDX

- Intel is developing ML-DSA based attestation support for TDX as part of PQC enablement, with ML-KEM key negotiation and ML-DSA attestation also in development for migration. ([OC3 Berlin 12th March 2026 Intel Security PQC Update](#), as of 2026-03-12, confidence: medium, source: grok)
- *New information about Intel's forward-looking PQC migration plans for TDX attestation (ML-DSA/ML-KEM), not previously recorded; complements rather than contradicts the existing note that current TDX API surface lacks PQC options.*

Security & network appliances

ADCs & load balancers (adc-load-balancers)

New developments (2)

A10

- A10 Networks announced the Thunder 3360S ADC mid-range hardware platform on September 2, 2026, explicitly designed as a PQC-ready platform for future quantum-resistant cryptography support. ([A10 ネットワークス、AIの活用や暗号化通信の増加に対応するミッドレンジADC「A10 Thunder 3360S ADC」を提供](#), as of 2026-09-03, confidence: medium, source: grok)
- *Introduces a new A10 hardware platform explicitly marketed as PQC-ready, not previously recorded.*

F5

- F5's ADSP Partner Program has validated 12 certificate lifecycle management (CLM) integrations with eight partners that automate certificate renewals and support rollout of PQC-ready (including hybrid) certificates on BIG-IP to accelerate PQC transition. ([Validated integrations, real outcomes: Inside the F5 ADSP Partner Program](#), as of 2026-08-27, confidence: medium, source: grok)
- *New specific detail on F5's partner ecosystem supporting PQC certificate rollout, not present in the existing knowledge base.*

Newly captured background -- older material, first time in the knowledge base (1)

A10

- A10 Networks participated in the PQC Taiwan Enterprise Post-Quantum Summit on or around August 13, 2026, to discuss enterprise PQC deployment strategies and risk mitigation. ([A10 Networks X post on PQC Taiwan summit](#), as of 2026-08-13, confidence: low, source: grok)
- *Genuinely new event-participation detail for A10, though source is blocked so confidence is kept low; does not supersede or conflict with anything existing.*

Routers, switches, SD-WAN (~~routers-switches-sdwan~~)

New developments (2)

IOS-XE

- Cisco IOS XE 26.2 adds ML-DSA signatures for IKEv2, enabling end-to-end quantum-safe site-to-site tunnels (key exchange from 26.1 plus authentication). ([Post-Quantum Key Exchange on Cisco Routers - IPsec Series, Part 9](#), as of 2026-09-01, confidence: high, source: grok)
- *Adds a new capability (ML-DSA authentication for IKEv2) not previously recorded, distinct from the existing ML-KEM key-exchange and image-signing claims.*

Junos

- Junos OS 26.2R1 adds support for Quantum Buffer for SSH on additional platforms including ACX, EX, MX, QFX, SRX series. ([Junos OS Release 26.2R1 Release Notes](#), as of 2026-08-31, confidence: high, source: grok)
- *Specifies the platform rollout (ACX, EX, MX, QFX, SRX) for the Quantum Buffer SSH feature, a detail not present in the earlier general claim about 25.4R1/26.2R1 support.*

VPN concentrators & WAN encryptors (~~vpn-concentrators-encryptors~~)

New developments (1)

VPN concentrator hardware

- Cisco IOS XE 26.2 on the 8000 Series Secure Routers (including C8235-G2) adds native ML-DSA signature support for IKEv2, enabling fully quantum-safe end-to-end site-to-site IPsec VPN tunnels with ML-KEM hybrid

key exchange and ML-DSA authentication. ([Post-Quantum Key Exchange on Cisco Routers - IPsec Series, Part 9](#), as of 2026-09-01, confidence: high, source: grok)

- *New capability beyond prior IOS-XE 26.1.1 ML-KEM-only support [d0004:5], adding ML-DSA signature authentication for a specific router line distinct from the FTD/ASA 11.0 (2027) roadmap in [d0001:6].*

Enterprise infrastructure & storage

Backup & storage systems (backup-storage)

New developments (1)

NetApp

- ONTAP 9.17.1 supports postquantum pre-shared keys (PPKs) for IPsec authentication. ([Learn about IP security on the ONTAP network](#), as of 2026-08-27, confidence: high, source: grok)
- *Introduces a distinct PQC feature (IPsec PPKs in ONTAP 9.17.1) not covered by the existing ONTAP SSL/ML-KEM claim.*

Superseded (1)

NetApp

- ONTAP 9.18.1 and later support ML-KEM, ML-DSA, and SLH-DSA for SSL when FIPS is disabled, with bidirectional negotiation. ([Which ONTAP feature may support to apply PQC algorithms?](#), as of 2026-04-01, confidence: high, source: grok)
- supersedes: "[d0002:3] ONTAP 9.18.1 introduces support for the ML-KEM, ML-DSA, and SLH-DSA post-quantum computing cryptographic algorithms for SSL, available when FIPS is disabled."
- *Adds new substantive detail (applies to 9.18.1 and later, and bidirectional negotiation mechanism) beyond the original claim.*

Peripherals & physical access (peripherals-physical-access)

New developments (2)

badge and physical access systems

- GSA is updating the FICAM architecture to support post-quantum cryptography with an emphasis on crypto agility for compatibility with existing systems, alongside expanding the FIPS 201 Evaluation Program / PACS lab to test quantum-resistant solutions. ([GSA Leads the Transition to Quantum-Resistant Technology | GSA](#), as of 2026-08-24, confidence: high, source: grok)
- *Adds a new architectural detail (FICAM modernization emphasizing crypto agility) beyond the previously recorded PACS lab expansion.*
- GSA's FICAM modernization includes an interagency working group whose first meeting occurred on August 12, 2026, with a phased timeline: discovery/planning 2026-2027, pilots 2027-2028, prioritized migration through 2030, post-quantum digital signatures for high-value systems by 2031, and full migration by 2035. ([US begins post-quantum upgrade of federal identity credentials | Biometric Update](#), as of 2026-08-27, confidence: high, source: grok)
- *Provides new concrete timeline and working-group details not present in any existing GSA-related claim.*

Newly captured background -- older material, first time in the knowledge base (1)

badge and physical access systems

- Swissbit iShield Key 2 integrates MIFARE DESFire EV3 credential technology (via NXP SmartMX3 P71D600 MCU) alongside FIDO2 for unified logical and physical access control in a single device, available as of June 2026. ([Swissbit combines FIDO and physical access control in single security key with NXP | Biometric Update](#), as of 2026-06-12, confidence: high, source: grok)
- *Describes a distinct additional credential-technology integration (MIFARE DESFire EV3/NXP) on the iShield Key 2 not covered by the earlier HID Seos integration claim.*

Disk & data-at-rest encryption (disk-and-data-encryption)

New developments (1)

SEDs/TCG Opal

- TCG released guidance (August 2026) defining requirements for PQC-ready TPMs, identifying PTP 1.07 as baseline and designating 'TCG PQC-ready TPM' (implements PTP 1.07) vs 'TCG PQC-upgradable TPM' categories. ([TCG Defines Requirements for PQC-Ready Trusted Platform Modules](#), as of 2026-08-24, confidence: high, source: grok)
- *This is a distinct TCG standard (TPM/PTP) from the Opal spec covered in the existing claim, and introduces a genuinely new PQC-readiness classification not previously recorded in this leaf.*

Governance, mandates & timelines

EU & UK guidance (eu-uk-guidance)

New developments (1)

NCSC

- As of September 2026, NCSC's PQC pilot under the Assured Cyber Security Consultancy scheme remains open for qualified consultancies (expression of interest closed due to high interest) and is scheduled to run until 31 March 2027, after which criteria and learning will be reviewed. ([Introduction | Post-quantum cryptography \(PQC\) pilot](#), as of 2026-09-02, confidence: high, source: grok)
- *Adds new status detail (expression-of-interest closure due to high demand and post-pilot review plans) not present in the earlier pilot announcement.*

Superseded (1)

EU/ETSI roadmap

- The EU Quantum Act proposal, originally expected in Q2 2026, has been delayed and is now tentatively planned for Q4 2026. ([EU's first quantum tech regulation delayed by six months](#), as of 2026-08-11, confidence: high, source: grok)
- supersedes: "[d0002:3] The EU is preparing a Quantum Act expected to be published in 2026, in the context of the June 2025 PQC roadmap."
- *Refines the vague 2026 expectation with a specific delay from Q2 to Q4 2026, a substantive scheduling update.*

Newly captured background -- older material, first time in the knowledge base (2)

EU/ETSI roadmap

- As of July 2026, ETSI CYBER QSC work programme includes multiple active draft items such as TR 104 317 on Quantum-safe signatures and PKI considerations (early draft July 2026), TS 104 145 on Quantum-Safe Enterprise Transport Security (final draft July 2026), and DMI/CYBER-QSC-0010 CYBER QSC Extended Roadmap (stable draft June 2026). ([Work Programme - Work Item Plan](#), as of 2026-07-28, confidence: high, source: grok)
- *Introduces specific new ETSI CYBER QSC work-item details (TR 104 317, TS 104 145, DMI/CYBER-QSC-0010) not previously recorded in the knowledge base.*

NCSC

- NCSC's July 2026 PQC migration workshop report highlights key themes including the need for executive sponsorship and board engagement through cost-of-delay messaging, peer benchmarks, and phased roadmaps; supply-chain readiness via early supplier engagement and clear PQC requirements; and industry transparency plus cross-sector collaboration to build national resilience. ([Post-quantum cryptography \(PQC\) migration workshop report](#), as of 2026-07-22, confidence: high, source: grok)
- *Adds substantial new detail on report themes (executive sponsorship, cost-of-delay messaging, peer benchmarks) beyond the previously recorded supply-chain summary.*

Procurement, contract language & audit (procurement-contract-language)

New developments (2)

RFP and supplier questionnaire language

- On August 28, 2026, CISA's lead for quantum security stated that agencies will modify contract language to require vendors to provide quantum-resistant products, demonstrate PQC compliance pathways, and support mechanisms such as cryptographic bills of materials. ([Quantum security 'drumbeat' to grow louder in federal procurement](#), as of 2026-08-28, confidence: high, source: grok)
- *New concrete statement from a CISA official confirming imminent contract-language changes requiring vendor CBOM support, beyond the general prediction in existing claims.*
- On August 28, 2026, the Department of Defense issued an RFI seeking industry input on software-defined encryption solutions providing packet-level cryptographic protection aligned with PQC migration timelines through December 31, 2031, with responses due September 27, 2026. ([DOD probes industry about software-defined encryption as U.S. scurries to get quantum-safe](#), as of 2026-08-28, confidence: high, source: grok)
- *New, distinct DoD procurement RFI on software-defined encryption with its own timeline, not previously recorded.*

Certification schemes (certification-schemes)

New developments (1)

FIPS 140-3 module transition timelines

- Sansec HSM Cryptographic Module received FIPS 140-3 Level 3 certification (certificate #5503) as the first domestic Chinese HSM supporting the full NIST PQC suite (ML-DSA, ML-KEM, SLH-DSA) alongside classical algorithms. ([三未信安取得FIPS 140-3 Level 3认证，全球化布局再落关键一子](#), as of 2026-08-27, confidence: high, source: grok)
- *New distinct vendor/product (Sansec, China's first) achieving FIPS 140-3 Level 3 with full PQC suite, not previously recorded — existing d0004:1 covers Crypto4A's HSM, a different product.*

Sector compliance regimes (sector-compliance)

Newly captured background -- older material, first time in the knowledge base (1)

PCI DSS

- PCI PTS HSM v5.0, published May 2026, introduces support for post-quantum cryptography considerations, new definitions covering post-quantum cryptography, strengthened cryptographic requirements (minimum 128-bit effective key strength for device-security keys, TDES no longer permitted for device security), and new evaluation modules for key-transfer, remote administration, and HSM solution security. ([PCI SSC Publishes PCI PTS HSM v5.0](#), as of 2026-05-18, confidence: high, source: grok)
- *New standard (PCI PTS HSM v5.0) and its PQC-related requirements are not present anywhere in the existing knowledge base, which only covers PCI DSS 4.0 and FedRAMP items.*

Hardware acceleration & silicon

CPU & ISA extensions (cpu-isa-extensions)

New developments (1)

vendor-optimized PQ assembly upstreamed into libraries

- wolfSSL retired its remaining dependency on liboqs in August 2026, completing the transition to fully native implementations of ML-KEM, ML-DSA, and other PQC algorithms with optimized paths for x86-64 and AArch64. ([wolfCrypt Retires liboqs as Native Post-Quantum Support Expands](#), as of 2026-08-19, confidence: high, source: grok)
- *New information about wolfSSL fully decoupling from liboqs and shipping native optimized PQC assembly, not previously recorded (prior claim only noted wolfSSL's own AVX2 ML-KEM assembly without addressing liboqs dependency status).*

Conflicting -- needs review (1)

RISC-V crypto extensions

- mlkem-native now includes native RISC-V (64-bit, RVV) backend alongside Arm Neon, x86_64 AVX2, and POWER VSX. ([GitHub - pq-code-package/mlkem-native](#), as of 2026-09-03, confidence: high, source: grok)
- conflicts with: "[d0001:23] all lattice-acceleration work for RISC-V found in current literature remains academic proposals ... rather than adopted ISA extensions"
- *Claims a production-library RISC-V RVV backend has been upstreamed, which would contradict the established assessment that RISC-V lattice acceleration remains academic-only, but the source URL was flagged as not found in citations, so it should not silently overwrite confirmed knowledge.*

Newly captured background -- older material, first time in the knowledge base (1)

vendor-optimized PQ assembly

- mldsa-native v2.0.0 released August 3, 2026 as first stable release with x86_64 AVX2 and AArch64 Neon assembly backends, SLOTHY super-optimizer integration, and completed formal verification (HOL-Light for assembly, CBMC for C). ([Releases · pq-code-package/mldsa-native](#), as of 2026-08-03, confidence: high, source: grok)
- *Reports a specific new stable release of mldsa-native with SLOTHY integration and completed formal verification, details not previously recorded (prior mention was only of liboqs 0.16.0 integrating mldsa-native, without this release/verification detail).*

Operating systems & endpoints

Linux distributions & kernel crypto (**linux-distros-kernel**)

Superseded (1)

SUSE

- SUSE Linux Enterprise Server 16.0 enables post-quantum key exchange algorithms by default in system-wide cryptographic policies for SSH connections, re-enabling the X25519 group to support hybrids such as mlkem768x25519-sha256 and sntrup761x25519-sha512. ([SUSE Linux Enterprise Server Release Notes | SUSE Linux Enterprise Server 16.0](#), as of 2026-08-19, confidence: medium, source: grok)
- supersedes: "[d0001:16] The OpenSSH 10.x hybrid ML-KEM/X25519 key exchange was, as of that SUSE analysis, not yet available in SUSE Linux Enterprise products."
- *SLES 16.0 now ships with PQC hybrid SSH key exchange enabled by default via updated crypto policies, superseding the earlier finding that this was not yet available.*

Apple platforms (**apple-platforms**)

Newly captured background -- older material, first time in the knowledge base (3)

iOS

- Quantum-secure TLS is enabled by default in URLSession and Network framework for all system services and apps on iOS 26, iPadOS 26, macOS 26, tvOS 26, and watchOS 26. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *New detail that quantum-secure TLS is on-by-default at the API level for system services and third-party apps, going beyond the earlier ClientHello-advertisement claim.*

macOS

- In macOS 26, the SSH protocol was upgraded with a quantum-secure encryption key exchange. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *New, specific, Apple-primary-sourced detail confirming SSH itself gained a quantum-secure KEX in macOS 26 -- distinct from and more precise than the earlier contested, PQ3-branded TLS/VPN/SSH claim.*

watchOS

- Quantum-secure encryption between iPhone (iOS 26+) and Apple Watch (watchOS 26+) uses additional ML-KEM-768 key exchanges in addition to elliptic-curve Diffie-Hellman. ([Apple Platform Security Guide](#), as of 2026-08-03, confidence: high, source: grok)
- *Genuinely new detail about a specific iPhone-Watch pairing protocol using hybrid ML-KEM-768 + ECDH, not previously recorded.*

Email & messaging

Secure messaging protocols (**secure-messaging-protocols**)

New developments (1)

MLS

- The Amortized PQ MLS Combiner draft was updated with publication date 2026-08-21. ([Amortized PQ MLS Combiner](#), as of 2026-08-21, confidence: high, source: grok)
- *Indicates a new draft revision published after the previously known 2026-04-09 version, a genuine update not previously recorded.*

Email security (email-security)

Newly captured background -- older material, first time in the knowledge base (1)

DirectTrust / Direct Secure Messaging

- DirectTrust describes Direct Secure Messaging as a secure communication transport for sensitive information over the open internet that utilizes digital certificates and a public key infrastructure to encrypt message contents so only the intended recipient can decrypt them, forming the certificate-anchored trust fabric any future PQ transition for the ecosystem would run through. ([directtrust.org — direct secure messaging](#), as of undated, confidence: low, source: ingest)
- *First claim in the base introducing DirectTrust/Direct Secure Messaging, a new subtopic area not previously covered.*

Financial messaging & sector rails

SWIFT & ISO 20022 (swift-iso20022)

Superseded (1)

ISO 20022 message signing

- SWIFT imposes no signature-size constraint for post-quantum signatures because authentication is applied out-of-band at the SWIFTNet messaging/transport layer, so neither the signature nor the verifying key travels in the Block-4 message text and every NIST PQC signature is size-compatible with SWIFT. ([Post-Quantum Cryptography Migration in Australian Real-Time Payment Infrastructure: A Monte Carlo Simulation Study of the New Payments Platform](#), as of 2026-05-04, confidence: medium, source: grok)
- supersedes: "[d0001:13] recommends Falcon-512 as primary PQC signature algorithm for SWIFT MT exposure since it is the only NIST PQC signature algorithm that fits within the 2,048-byte SWIFT MT message limit"
- *New claim from the same study corrects the underlying premise that drove the Falcon-512-for-SWIFT recommendation, showing SWIFTNet authentication is out-of-band and not subject to the in-band size limit.*

Newly captured background -- older material, first time in the knowledge base (2)

correspondent-banking crypto requirements

- ML-DSA-65 with ML-KEM-768 is recommended as the default PQC algorithms across NPP and SWIFT routes in correspondent-banking payment infrastructure. ([Post-Quantum Cryptography Migration in Australian Real-Time Payment Infrastructure: A Monte Carlo Simulation Study of the New Payments Platform](#), as of 2026-05-04, confidence: medium, source: grok)
- *Introduces a specific concrete default algorithm pairing not previously stated (prior claim only said 'security level per NIST guidance').*
- Falcon-512 is recommended where genuinely in-band, size-constrained carriage is required in payment infrastructure including SWIFT. ([Post-Quantum Cryptography Migration in Australian Real-Time Payment Infrastructure: A Monte Carlo Simulation Study of the New Payments Platform](#), as of 2026-05-04, confidence:

medium, source: grok)

- *Adds a new, narrower scope condition for Falcon-512 use that wasn't previously captured (limited to genuinely constrained in-band carriage rather than all SWIFT MT exposure).*

Financial standards bodies (~~financial-standards-bodies~~)

Newly captured background -- older material, first time in the knowledge base (2)

ASC X9 PQC work

- X9 published standard X9.150 'Merchant-Presented QR Codes For Secure Payment' on July 14, 2026, which states that payment QR codes and associated information are digitally signed using public key cryptography supported by the X9 Financial PKI, helping usher in the next generation of post-quantum cryptography signatures. ([New X9 Standard for Secure, Interoperable QR Code Payments Will Accelerate U.S. Use of Instant Payments](#), as of 2026-07-14, confidence: high, source: grok)
- *A newly published X9 standard (X9.150) referencing PQC signatures is not present anywhere in the existing knowledge base.*

EMVCo

- EMVCo released the second of three position papers, titled 'Security Position Statement: Update on Quantum Computing and EMV Chip Cryptography', by Q1 2026. ([EMVCo Newsletter: Q1 2026](#), as of 2026-04-10, confidence: high, source: grok)
- *This confirms actual release of a paper that was previously only described as planned/in development in d0002:2, marking a change in status from planned to published.*

Enterprise applications & delivery

ERP & business platforms (~~erp-business-platforms~~)

Superseded (1)

Oracle Database

- Official Oracle Database documentation as of August 2026 confirms that PQC algorithms (ML-KEM, ML-DSA, SLH-DSA) are not available in FIPS 140-2 mode regardless of cryptographic provider, available only in non-FIPS or FIPS 140-3 mode with the next-generation provider. ([Post-Quantum Cryptography in Oracle Database](#), as of 2026-08-25, confidence: high, source: grok)
- supersedes: "[d0004:3] a August 2026 Oracle blog states the new cryptographic provider in Oracle Database 19.32 supports non-FIPS, FIPS 140-2, and FIPS 140-3 modes"
- *Official Oracle documentation (higher authority than the blog post) clarifies that PQC remains unavailable in FIPS 140-2 mode regardless of provider, resolving the prior unresolved conflict in favor of the original claim.*

Newly captured background -- older material, first time in the knowledge base (1)

SAP

- SAP updated its SAP HANA cloud system with PQC support for TLS connections using hybrid key exchange combining classical and quantum-resistant algorithms. ([Microsoft and SAP accelerate quantum security for enterprise](#), as of 2026-07-06, confidence: medium, source: grok)
- *Introduces PQC/hybrid TLS support specifically for SAP HANA Cloud, a product not covered by existing SAP claims which address BTP, NetWeaver ABAP, and the Common Cryptographic Library.*

VDI & remote access (**vdi-remote-access**)

Conflicting -- needs review (1)

Citrix

- NetScaler 14.1 build 51.72 and later include hybrid post-quantum key exchange through the X25519_MLKEM768 curve; both default SSL profiles bind this curve, but TLS 1.3 must be enabled for negotiation, and SSL Labs now reports PQC key exchange support status. ([Networking SSL/TLS Best Practices \(Q3 2026 Edition\)](#), as of 2026-08-25, confidence: high, source: grok)
- conflicts with: "Citrix docs (Support for Hybrid Post-Quantum Cryptography on the front end): 'the feature is available only for enhanced SSL profiles' — this claim instead states both default SSL profiles bind the curve automatically"
- *This claim's assertion that default SSL profiles automatically bind X25519_MLKEM768 contradicts the official Citrix documentation's statement that the feature requires an enhanced SSL profile, and it's unclear which supersedes the other.*

Newly captured background -- older material, first time in the knowledge base (1)

Citrix

- To enable hybrid PQC (X25519_MLKEM768) on NetScaler, configure an enhanced SSL profile by binding the X25519_MLKEM768 elliptic curve (via CLI or GUI), with the note that the feature is available only for enhanced SSL profiles. ([Support for Hybrid Post-Quantum Cryptography on the front end](#), as of 2025-09-27, confidence: high, source: grok)
- *Adds a new configuration detail (enhanced-SSL-profile-only requirement, CLI/GUI binding steps) not previously captured for NetScaler's PQC support.*

Containers, orchestration, service mesh

Container runtimes (**container-runtimes**)

Newly captured background -- older material, first time in the knowledge base (1)

Docker

- Docker Engine and Docker CLI releases (including the 29.x series) are built with Go 1.26.x toolchains, enabling X25519MLKEM768 hybrid post-quantum key exchange by default in crypto/tls for outbound TLS connections such as to registries over HTTPS, provided no explicit CurvePreferences override. ([Releases · moby/moby](#), as of 2026-08-05, confidence: high, source: grok)
- *Distinct from prior Mirantis Container Runtime and containerd Go-1.26 claims, this newly identifies Docker Engine/CLI itself as inheriting default hybrid PQ TLS via its Go 1.26 toolchain.*

Service mesh & workload identity (**mesh-workload-identity**)

Newly captured background -- older material, first time in the knowledge base (3)

Consul

- As of July 16, 2026, an open GitHub issue requests that Consul add support for post-quantum algorithms such as X25519MLKEM768 for TLS key exchange, noting that Go's crypto/tls has supported them since version

1.24 but documentation is unclear. ([Support of post-quantum algorithms · Issue #23738 · hashicorp/consul](#), as of 2026-07-16, confidence: medium, source: grok)

- *New, distinct issue about Consul's own TLS stack lacking PQC support, separate from the previously known stale upstream Envoy issue.*

Istio

- As of May 16, 2025, an open Istio GitHub issue requests extending PQC support (X25519MLKEM768) from gateways to mesh-internal traffic including sidecars, waypoints, ztunnel, and xDS, suggesting a new 'pqc' compliance policy. ([Feature request: PQC for mesh-internal traffic · Issue #56330 · istio/istio](#), as of 2025-05-16, confidence: medium, source: grok)
- *Adds new detail about a pre-1.27 feature request explicitly calling out sidecars and waypoints, coverage not confirmed by the existing 1.27 compliance-policy claim.*

SPIFFE/SPIRE

- SPIRE 1.14.3 (March 18, 2026) updated the RequirePQKEM TLS policy to use the standardized X25519MLKEM768 instead of the draft x25519Kyber768Draft00. ([spire/CHANGELOG.md at main · spiffe/spire](#), as of 2026-03-18, confidence: high, source: grok)
- *New fact about an earlier SPIRE release (1.14.3) migrating from draft to standardized KEM naming, not previously recorded.*